Inference for Simple Linear Regression

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From Estimation to Inference

Where We Left Off

- Fitted the simple linear regression model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ by least squares
- Estimated $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
- Estimated $\hat{\sigma}^2 = \mathrm{MS}_{\mathrm{Res}} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n-2} = \frac{\sum_{i=1}^{n} \hat{\varepsilon}_i^2}{n-2} = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n-2}$

If we drew a different sample, would $\hat{\beta}_1$ come out the same?

Why Do We Need Inference?

Recall the rocket propellant example from the previous lecture.

- $\hat{\beta}_0$ and $\hat{\beta}_1$ are computed from *one* particular sample of 20 propellant batches
- A different sample of 20 batches would give slightly different numbers — the estimates are random variables
- Point estimates alone cannot answer questions like:
  - Is the apparent age effect on shear strength *real*, or could it be sampling noise?
  - How precisely do we know β_1 — within 1 psi/week, or within 50?
  - How much should we trust a prediction made from this line?

Inference (hypothesis tests and confidence intervals) turns a single fitted line into statements accompanied with some level of uncertainty.

Rocket Propellant Data: Is the Age Effect Real?

- Last time we fit $\hat{\beta}_1 = -37.15$ psi/week for the rocket-propellant data
  Question: if propellant age (in weeks) truly had *no* effect on shear strength (meaning that $\beta_1 = 0$), could a sample of 20 batches of propellant still produce a slope this negative just by chance?

- Answering this requires knowing how $\hat{\beta}_1$ varies from sample to sample, which is called its **sampling distribution**.

Sampling Distribution of the OLS Estimators

Sampling Distribution of $\hat{\beta}_1$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \sum_{i=1}^{n} c_i Y_i, \qquad c_i = \frac{x_i - \bar{x}}{S_{xx}}$$

- $\hat{\beta}_1$ is a linear combination of independent normal random variables, therefore, $\hat{\beta}_1$ is normally distributed

Further,

$$\mathrm{E}[\hat{\beta}_1] = \beta_1 \qquad \text{(unbiased)}$$

$$\mathrm{Var}(\hat{\beta}_1) = \sigma^2 \sum_i c_i^2 = \frac{\sigma^2}{S_{xx}}$$

Therefore:

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

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Sampling Distribution $\hat{\beta}_0$

Since $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$ and

$$\mathrm{E}[\hat{\beta}_0] = \beta_0 \qquad \text{(unbiased)}$$

$$\mathrm{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

Therefore:

$$\hat{\beta}_0 \sim N\left(\beta_0,\ \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \right)$$

Estimating the Error Variance

σ^2 Is Unknown — Estimate It from the Residuals

$$\hat{\sigma}^2 = \mathrm{MS}_{\mathrm{Res}} = \frac{\mathrm{SS}_{\mathrm{Res}}}{n-2}, \qquad \mathrm{SS}_{\mathrm{Res}} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

- Two degrees of freedom are used up estimating β_0, β_1
- $\mathrm{MS}_{\mathrm{Res}}$ is unbiased: $\mathrm{E}[\mathrm{MS}_{\mathrm{Res}}] = \sigma^2$
- Key distributional result:

$$\frac{(n-2)\mathrm{MS}_{\mathrm{Res}}}{\sigma^2} \sim \chi^2_{n-2}$$

Standard Errors

- Replace the unknown σ^2 by $\mathrm{MS}_{\mathrm{Res}}$ in the variance formulas

$$\mathrm{se}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} = \sqrt{\frac{\mathrm{MS}_{\mathrm{Res}}}{S_{xx}}}$$

$$\mathrm{se}(\hat{\beta}_0) = \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)} = \sqrt{\mathrm{MS}_{\mathrm{Res}} \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)}$$

- Because σ^2 is estimated, the reference distribution moves from normal to t_{n-2}

t-Test for the Slope

Test Statistic for β_1

$$T_0 = \frac{\hat{\beta}_1 - \beta_{1,0}}{\text{se}(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\text{MS}_{\text{Res}}/S_{xx}}} \sim t_{n-2} \quad \text{under } H_0$$

- $H_0 : \beta_1 = \beta_{1,0}$ vs. $H_1 : \beta_1 \neq \beta_{1,0}$ (or one-sided)
- $n - 2$ degrees of freedom — one for each parameter already estimated

Decision Rule

- Two-sided test at level α: reject H_0 if $|t_0| > t_{\alpha/2,\, n-2}$
- Equivalently, reject if the p-value $< \alpha$
- Most common special case: $\beta_{1,0} = 0$
  - Tests whether x has *any* linear relationship with y
  - Called the **test for significance of regression**

Running Example: The Rocket-Propellant Data

- Recall the rocket-propellant data from last lecture: age of propellant x (weeks) and shear strength y (psi), $n = 20$

$\bar{x}$	$\bar{y}$	S_{xx}	S_{xy}	$\mathrm{SS_T}$
13.3625	2131.36	1106.56	$-41{,}112.65$	$1{,}693{,}737.60$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-41{,}112.65}{1106.56} = -37.1536$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 2131.36 - (-37.1536)(13.3625) = 2627.82$$

Testing $H_0 : \beta_1 = 0$ for the Example

$$\begin{aligned}
\mathrm{SS_R} &= \hat{\beta}_1 S_{xy} = (-37.1536)(-41{,}112.65) = 1{,}527{,}482.95 \\
\mathrm{SS_{Res}} &= \mathrm{SS_T} - \mathrm{SS_R} = 1{,}693{,}737.60 - 1{,}527{,}482.95 = 166{,}254.65 \\
\mathrm{MS_{Res}} &= \frac{\mathrm{SS_{Res}}}{n-2} = \frac{166{,}254.65}{18} = 9236.37 \\
\mathrm{se}(\hat{\beta}_1) &= \sqrt{\frac{9236.37}{1106.56}} = 2.889 \\
t_0 &= \frac{-37.1536 - 0}{2.889} = -12.86
\end{aligned}$$

- $t_{0.025,18} = 2.101$; since $|t_0| = 12.86 \gg 2.101$, reject H_0
- Strong evidence that the age of the propellant is associated with shear strength

t-Test for the Intercept

Test Statistic for β_0

$$T_0 = \frac{\hat{\beta}_0 - \beta_{0,0}}{\mathrm{se}(\hat{\beta}_0)} \sim t_{n-2} \quad \text{under } H_0 : \beta_0 = \beta_{0,0}$$

- Same mechanics as the slope test, different standard error

  Question: for the rocket-propellant data, is $\beta_0 = 0$ an interesting hypothesis?

Confidence Intervals for the Slope and Intercept

CI for β_1

$$\hat{\beta}_1 \pm t_{\alpha/2,\, n-2}\ \operatorname{se}(\hat{\beta}_1)$$

- A $100(1-\alpha)\%$ CI: the range of slopes plausible given the data
- Narrower when S_{xx} is large or $\mathrm{MS}_{\mathrm{Res}}$ is small

CI for β_1 — Example

$$\hat{\beta}_1 \pm t_{0.025,18}\ \mathrm{se}(\hat{\beta}_1)$$
$$-37.15 \pm 2.101(2.889)$$
$$-37.15 \pm 6.07$$

$$95\%\ \text{CI}: \quad (-43.22,\ -31.08)$$

- Zero is nowhere near this interval — consistent with rejecting $H_0 : \beta_1 = 0$
- We are 95% confident that, on average, shear strength drops somewhere between 31 and 43 psi per additional week of age

CI for β_0

$$\hat{\beta}_0 \pm t_{\alpha/2,\, n-2}\ \ \mathrm{se}(\hat{\beta}_0)$$

- Same construction, using $\mathrm{se}(\hat{\beta}_0) = \sqrt{\mathrm{MS}_{\mathrm{Res}}(1/n + \bar{x}^2/S_{xx})}$
- Interpretation only meaningful if $x = 0$ falls in or near the range of the data

Partitioning Variability: The ANOVA Identity

Splitting Total Variation

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\mathrm{SS_T}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\mathrm{SS_R}} + \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{\mathrm{SS_{Res}}}$$

- $\mathrm{SS_T}$: total variability in the response
- $\mathrm{SS_R}$: variability *explained* by the regression on x
- $\mathrm{SS_{Res}}$: variability left *unexplained* (residual)

Degrees of Freedom Split the Same Way

$$\underbrace{n-1}_{\mathrm{SS_T}} = \underbrace{1}_{\mathrm{SS_R}} + \underbrace{n-2}_{\mathrm{SS_{Res}}}$$

- One degree of freedom for the regression — a single slope parameter
- $n-2$ left for error, matching what was used for $\mathrm{MS_{Res}}$

The ANOVA Table

Source	df	SS	MS	F_0
Regression	1	$\mathrm{SS_R}$	$\mathrm{MS_R} = \mathrm{SS_R}/1$	$\mathrm{MS_R}/\mathrm{MS_{Res}}$
Residual (Error)	$n-2$	$\mathrm{SS_{Res}}$	$\mathrm{MS_{Res}} = \mathrm{SS_{Res}}/(n-2)$	
Total	$n-1$	$\mathrm{SS_T}$		

- Rows sum: the df's add to $n-1$, the SS's add to $\mathrm{SS_T}$

ANOVA Table for the Example

Source	df	SS	MS	F_0
Regression	1	1,527,482.95	1,527,482.95	165.38
Residual	18	166,254.65	9236.37	
Total	19	1,693,737.60		

- $\mathrm{SS_R} = \hat{\beta}_1 S_{xy} = (-37.1536)(-41{,}112.65) = 1{,}527{,}482.95$
- $F_0 = 1{,}527{,}482.95 / 9236.37 = 165.38$

The F-Test for Significance of Regression

F-Test

$$F_0 = \frac{\mathrm{MS_R}}{\mathrm{MS_{Res}}} \sim F_{1,\,n-2} \quad \text{under } H_0 : \beta_1 = 0$$

- Reject H_0 if $F_0 > F_{\alpha,\,1,\,n-2}$
- For the example: $F_{0.05,1,18} = 4.41$; since $F_0 = 165.38 \gg 4.41$, reject H_0

F-Test vs. t-Test

Question: in simple linear regression, how do the t-test and F-test for $H_0 : \beta_1 = 0$ relate?

- Check against the numbers above: $t_0^2 = (-12.86)^2 = 165.38 = F_0$

Coefficient of Determination

R^2: How Much Variation Is Explained

$$R^2 = \frac{\mathrm{SS_R}}{\mathrm{SS_T}} = 1 - \frac{\mathrm{SS_{Res}}}{\mathrm{SS_T}}, \qquad 0 \le R^2 \le 1$$

- Proportion of the total variability in y explained by the regression on x
- For the example: $R^2 = 1{,}527{,}482.95/1{,}693{,}737.60 = 0.902$ — about 90% of the variability in shear strength is explained by the age of the propellant

R^2 and the Correlation Coefficient

- In simple linear regression only:

$$R^2 = r^2$$

- $r = \text{sign}(\hat{\beta}_1)\sqrt{R^2} = -\sqrt{0.902} = -0.950$
- This equivalence holds only for SLR — it does *not* hold once there is more than one predictor

Prediction

Two Different Questions

- Fix a value x_0. Two distinct targets:
  - the *mean* response $\mathrm{E}[Y \mid x_0] = \beta_0 + \beta_1 x_0$
  - a *single new* observation $Y_0 = \beta_0 + \beta_1 x_0 + \varepsilon_0$
- Both are estimated by the same point, $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$
- But they do *not* share the same uncertainty

Mean Response vs. New Observation

Suppose $x_0 = 10$ weeks.

- **Mean response:** the *average* shear strength we would see if we could test an unlimited number of 10-week-old batches — a single fixed (if unknown) number, $\beta_0 + \beta_1(10)$
- **New observation:** the shear strength of *one specific* 10-week-old batch about to be tested — it carries its own error term ε_0 on top of everything else

  Question: which should be wider — an interval for the average of many batches, or an interval for one particular batch?

Why the Prediction Interval Is Wider

The prediction error is $Y_0 - \hat{y}_0$, and Y_0 is independent of the data used to fit $\hat{y}_0$:

$$\begin{aligned}\operatorname{Var}(Y_0 - \hat{y}_0) &= \operatorname{Var}(Y_0) + \operatorname{Var}(\hat{y}_0) \\ &= \sigma^2 \;+\; \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right) \\ &= \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)\end{aligned}$$

- The mean-response error is just $\hat{y}_0 - \mathrm{E}[Y \mid x_0]$, with variance $\sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$ — no extra “1”
- That extra σ^2 is ε_0’s own variance, and it cannot be estimated away, no matter how large n is

The Two Interval Formulas

Mean response

$$\hat{y}_0 \pm t_{\alpha/2,\,n-2}\sqrt{\mathrm{MS}_{\mathrm{Res}}\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$$

New observation

$$\hat{y}_0 \pm t_{\alpha/2,\,n-2}\sqrt{\mathrm{MS}_{\mathrm{Res}}\left(1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$$

- Same centre, same critical value — only the "$+1$" inside the square root differs
- Both are narrowest at $x_0 = \bar{x}$ and widen as x_0 moves away from $\bar{x}$

CI and PI at $x_0 = 10$ Weeks

$$\hat{y}_0 = 2627.82 + (-37.1536)(10) = 2256.29$$

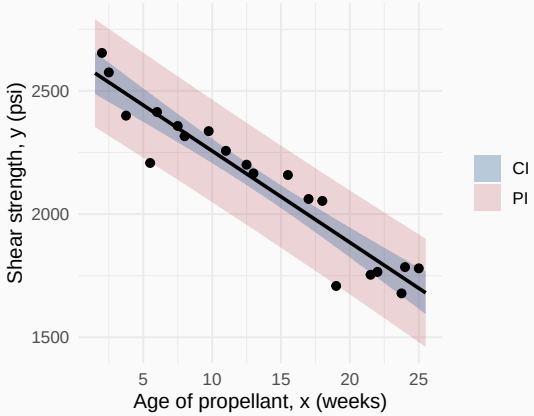
$$\text{se(mean)} = \sqrt{9236.37\left(\tfrac{1}{20} + \tfrac{(10-13.3625)^2}{1106.56}\right)} = 23.58$$

$$\text{se(pred)} = \sqrt{9236.37\left(1 + \tfrac{1}{20} + \tfrac{(10-13.3625)^2}{1106.56}\right)} = 98.96$$

- 95% CI for the mean response:
  $2256.29 \pm 2.101(23.58) = (2206.74,\ 2305.84)$
- 95% PI for a new observation:
  $2256.29 \pm 2.101(98.96) = (2048.38,\ 2464.20)$

CI Band vs. PI Band

- Both bands are narrowest at $x = \bar{x}$ and flare outward
- The PI band is always the wider of the two, everywhere
- Neither band should be trusted far outside the range of the observed x_i



Extrapolation Warning

- Both intervals are derived under the model fit *within* the observed range of x
- Predicting far outside that range assumes the linear relationship continues to hold — often unjustified

  The rocket-propellant data covers ages x from 2 to 25 weeks. How much would you trust a prediction at $x_0 = 100$ weeks?

Questions?